

1. A 67-year-old white man comes to the physician because his wife is concerned about a freckle on the back of his neck. He says he has had the freckle for many years, but he has noticed that it has grown during the past 6 months. His brother died of liver cancer at the age of 50 years. Examination shows a 3-cm lesion at the base of the skull. A photograph of the lesion is shown. The remainder of the examination shows no abnormalities. Which of the following is the most appropriate next step in management?

- ☐ A) Follow-up examination in 6 months
- ☐ B) Topical application of retinoic acid
- ☐ C) Full-thickness biopsy
- ☐ D) Shave biopsy
- ☐ E) Cryoablation



2. A 24-year-old woman has had intermittent fever, malaise, and a generalized rash for 2 weeks. Her temperature is 37.2°C (99°F), pulse is 80/min, respirations are 14/min, and blood pressure is 124/76 mm Hg. There is a generalized erythematous maculopapular rash over the palms and soles. She has patchy alopecia of the scalp and eyebrows. Hematocrit is 36%, and leukocyte count is 9000/mm³ (70% segmented neutrophils and 30% lymphocytes). She is given intravenous aqueous penicillin G. Two hours later, she has severe chills, headache, tachycardia, flushing, and hyperventilation. Her temperature is 39.4°C (103°F), and blood pressure is 90/60 mm Hg. Which of the following is the most likely cause of her acute condition?

- ☐ A) Chronic meningococemia
- ☐ B) Jarisch-Herxheimer reaction
- ☐ C) Rocky Mountain spotted fever
- ☐ D) Systemic lupus erythematosus
- ☐ E) Toxic shock syndrome



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3. A 53-year-old man notices a lump in his neck. Examination shows a hard 2-cm nodule in the lower left pole of the thyroid gland. Needle biopsy shows papillary carcinoma, and he undergoes total thyroidectomy. Five days later, he reports facial numbness and tingling, muscle pain, and hoarseness. Physical examination shows laryngeal stridor and a positive Chvostek sign. Which of the following is the most appropriate management?

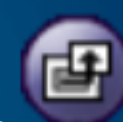
- ☐ A) Hydration with 0.9% isotonic saline intravenously
- ☐ B) Administration of calcium gluconate intravenously
- ☐ C) Administration of glucocorticoids
- ☐ D) Administration of 50,000 U of vitamin D
- ☐ E) Endotracheal intubation



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4. A 77-year-old man comes to the physician because of a 3-month history of intermittent mild abdominal pain. He also has had rectal pain with bowel movements and occasionally notices traces of red blood in his stool. He has type 2 diabetes mellitus, benign prostatic hyperplasia, and hypertension. Current medications include insulin, tamsulosin, lisinopril, and hydrochlorothiazide; verapamil was added 4 months ago. A colonoscopy performed 1 year ago showed no abnormalities. He is not in distress. His blood pressure is 108/72 mm Hg. Examination shows moist mucous membranes. The abdomen is slightly distended and nontender. The prostate is enlarged, and there is an external hemorrhoid. Rectal examination shows hard stool in the rectal vault. Sensation to light touch and vibration is intact over the distal lower extremities. An x-ray of the abdomen shows multiple air-fluid levels and stool throughout the bowel. Which of the following is the most likely cause of this patient's current symptoms?

- ☐ A) Adverse effect of verapamil
- ☐ B) Colon cancer
- ☐ C) Dehydration
- ☐ D) Diabetic neuropathy of the colon
- ☐ E) Metastatic prostate cancer



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5. A 52-year-old woman with hypertension and type 2 diabetes mellitus is admitted to the hospital because of a 2-day history of increasing confusion. She is left-hand dominant. Her medications are lisinopril and metformin. On admission, she is awake and alert. Her temperature is 38.3°C (101°F), pulse is 100/min, and blood pressure is 130/70 mm Hg. On examination, muscle strength is 5/5 bilaterally. Sensation to light touch and pinprick is intact. Coordination is normal. She cannot recall a name and address after 5 minutes. A T2-weighted MRI of the brain is shown. Which of the following is the most appropriate pharmacotherapy?

- ☐ A) Acyclovir
- ☐ B) Ceftriaxone
- ☐ C) Dexamethasone
- ☐ D) Low-molecular-weight heparin
- ☐ E) Mannitol
- ☐ F) Pyrimethamine-sulfadiazine



6. A previously healthy 52-year-old man comes to the physician because of a 6-month history of cough productive of white sputum. He has had no chest pain. He has smoked one pack of cigarettes daily for 25 years. His pulse is 80/min and regular, respirations are 18/min, and blood pressure is 136/86 mm Hg. Breath sounds are decreased throughout with a prolonged expiratory phase. Cardiac examination shows no abnormalities. Spirometry shows an $FEV_1:FVC$ ratio of 0.65. Which of the following is the most likely explanation for these findings?

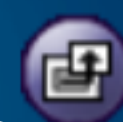
- ☐ A) Alveolar destruction and fibrosis
- ☐ B) Diffuse infiltrative disease of the lung parenchyma
- ☐ C) Diffuse thickening of the pleura
- ☐ D) Histamine-mediated bronchospasm and edema
- ☐ E) Mass effect in peripheral lung parenchyma



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7. A previously healthy 22-year-old man is brought to the emergency department 1 hour after he was hit in his right flank by a hockey stick. He reports no blood in his urine. He rates his pain as a 6 on a 10-point scale. His pulse is 100/min, and blood pressure is 140/90 mm Hg. Examination shows moderate tenderness to palpation over the right flank. His hematocrit is 42%. Urinalysis shows no abnormalities. Which of the following is the most appropriate next step in diagnosis?

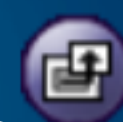
- ☐ A) Diagnostic laparoscopy
- ☐ B) Renal arteriography
- ☐ C) Retrograde urethrography
- ☐ D) Spiral CT scan of the abdomen
- ☐ E) No further testing is indicated



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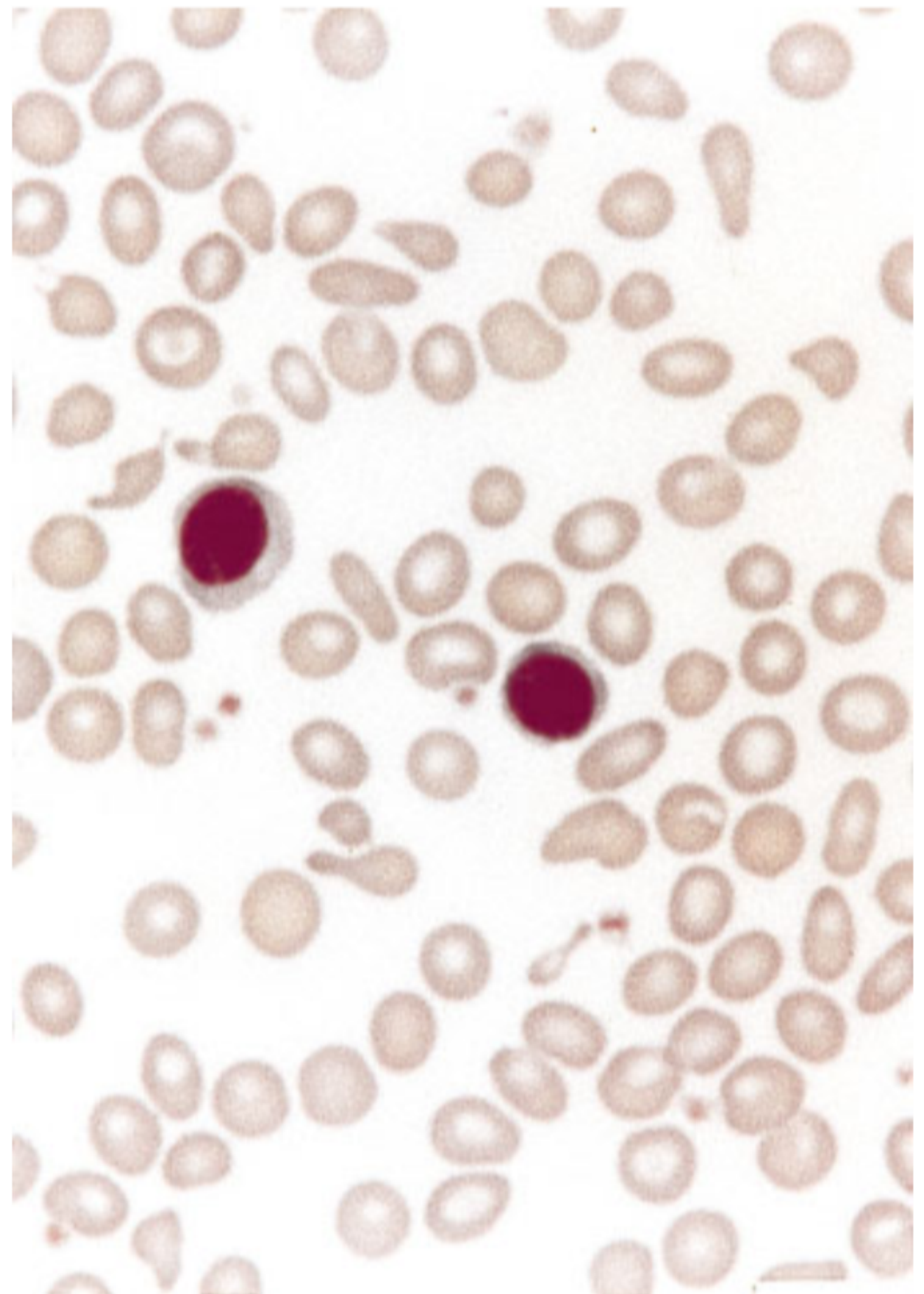
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8. A 24-year-old woman, gravida 1, para 1, comes to the physician because of a 3-month history of fatigue. Menses occur at regular intervals with no excessive bleeding. She takes no medications or vitamin supplements. Examination shows no abnormalities except for pale conjunctivae. Her hemoglobin concentration is 9 g/dL. A blood smear is shown. This patient most likely has a deficiency of which of the following?

- ☐ A) Folic acid
- ☐ B) Iron
- ☐ C) Niacin
- ☐ D) Vitamin B₁ (thiamine)
- ☐ E) Vitamin B₆



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9. A 62-year-old woman comes to the physician 8 days after she had loss of consciousness while combing her hair. She says she felt light-headed before she passed out. During the past 3 months, she has had two similar episodes and has noticed that her left arm tires easily. She is left-handed. She has hypertension treated with hydrochlorothiazide and atenolol. She has smoked one pack of cigarettes daily for 40 years. Her pulse is 76/min and regular, and blood pressure is 148/92 mm Hg in the right arm. Heart sounds are normal. The radial and brachial pulses are absent on the left and present on the right. Pedal pulses are decreased bilaterally. Muscle strength and sensation are normal. Which of the following is the most likely cause of this patient's symptoms?

- ☐ A) Coarctation of the aorta
- ☐ B) Left cervical rib
- ☐ C) Raynaud disease
- ☐ D) Subclavian steal syndrome
- ☐ E) Thromboangiitis obliterans
- ☐ F) Transient ischemic attack



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10. A 62-year-old woman comes to the physician for a follow-up examination. Her blood pressure was 140/90 mm Hg 3 months ago and 138/92 mm Hg 6 months ago. She has a 5-year history of type 2 diabetes mellitus well controlled with metformin. Her only other medication is aspirin. She has tried to limit her sodium intake to 2 g daily and reports moderate success. Her pulse is 80/min, and blood pressure is 145/85 mm Hg. Funduscopic examination shows no abnormalities. An S_4 is heard. Urinalysis shows:

Glucose	negative
Protein	1+
RBC	none
WBC	none

An ECG shows no abnormalities. Which of the following is the most appropriate next step in management?

- ☐ A) Schedule a follow-up visit in 3 months
- ☐ B) Decrease her sodium intake to 1 g daily
- ☐ C) Begin hydrochlorothiazide therapy
- ☐ D) Begin lisinopril therapy
- ☐ E) Begin metoprolol therapy



11. A 35-year-old man has had recurrent episodes of dysuria and pelvic pain over the past 2 years. He is sexually active. Examination shows a tender prostate. Urinalysis shows 2–5 WBC/hpf. A urine culture is negative. Which of the following is the most likely cause?

- ☐ A) Chancroid
- ☐ B) *Chlamydia trachomatis*
- ☐ C) Herpes simplex
- ☐ D) Syphilis
- ☐ E) *Trichomonas vaginalis*



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12. A 32-year-old man comes to the physician because of a 4-week history of moderate numbness and weakness of his left hand. There is no history of trauma to the hand. He has hypertension treated with enalapril and type 1 diabetes mellitus treated with insulin. He is 173 cm (5 ft 8 in) tall and weighs 63 kg (140 lb); BMI is 21 kg/m². His temperature is 37°C (98.6°F), pulse is 84/min, respirations are 16/min, and blood pressure is 146/88 mm Hg. Examination shows atrophy of the hypothenar muscles. The middle, ring, and small fingers are held in flexion; there is weakness of abduction of the small finger. The patient cannot prevent paper from being pulled through his fingers. All other intrinsic hand muscles are normal. Deep tendon reflexes are 1+ at the left brachioradialis, biceps, and triceps muscles. Sensation to pinprick is decreased over the palmar aspect of the left ring and small fingers and the hypothenar region of the hand. The remainder of the examination shows no other abnormalities. Which of the following is the most likely cause of these findings?

- ☐ A) Cervical syringomyelia
- ☐ B) C8 nerve root compression
- ☐ C) Lesion of the lower trunk of the brachial plexus
- ☐ D) Lesion of the medial cord of the brachial plexus
- ☐ E) Ulnar nerve compression



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13. A 77-year-old woman comes to the physician because of a 3-week history of light-headedness. She states that she feels as if she is going to pass out when she gets out of bed in the morning. She has not had chest pain, shortness of breath, or palpitations. She has hypertension and hypothyroidism. Current medications include hydrochlorothiazide, enalapril, and levothyroxine. She appears fatigued. She is 183 cm (6 ft) tall and weighs 86 kg (190 lb); BMI is 26 kg/m². Her temperature is 37.2°C (99°F), pulse is 64/min and irregularly irregular, respirations are 16/min, and blood pressure is 150/90 mm Hg while supine and 135/85 mm Hg while standing. Examination shows dry skin. The thyroid gland is symmetrically enlarged, nodular, and nontender. Jugular venous pulsations are present 3 cm above the sternal angle and decrease by 2 cm with inspiration. The lungs are clear to auscultation. A grade 1/6 systolic murmur is heard best at the left sternal border without radiation. There is trace pitting edema of the ankles. Neurologic examination shows no focal findings. Results of a complete blood count are within the reference range. Her serum potassium concentration is 3.4 mEq/L, and serum thyroid-stimulating hormone concentration is 1.5 μ U/mL. An ECG shows atrial fibrillation with a ventricular rate of 64/min and is otherwise unchanged from an ECG 1 year ago. Which of the following is the most appropriate next step to determine the cause of this patient's light-headedness?

- ☐ A) Measurement of serum free thyroxine concentration
- ☐ B) Serial measurements of serum creatine kinase activity
- ☐ C) Thyrotropin-releasing hormone stimulation test
- ☐ D) Ambulatory ECG monitoring
- ☐ E) Exercise stress test



14. A 52-year-old woman has decreased urine output 6 hours after admission to the hospital for treatment of a 3-day history of nausea and vomiting. She has type 2 diabetes mellitus and peripheral neuropathy. Her outpatient medications were glipizide, atorvastatin, lisinopril, and 81-mg aspirin; all medications were held on admission. Her blood pressure was 82/40 mm Hg on admission. After beginning intravenous administration of fluids, her blood pressure increased to 110/70 mm Hg. On examination now, the abdomen is soft, and there is diffuse mild tenderness with no guarding or rebound. Bowel sounds are decreased. There is decreased sensation to light touch over the feet. Laboratory studies show:

	On Admission	Now
Fingerstick blood glucose	4+	4+
Serum		
Urea nitrogen (mg/dL)	38	54
Creatinine (mg/dL)	1.8	2.6
Urine		
Specific gravity	1.028	1.011
RBC (/hpf)	0–2	0–2
Casts	None	coarse granular

Which of the following is the most likely explanation for this patient's decreased urine output?

- ☐ A) Acute glomerulonephritis
- ☐ B) Acute interstitial nephritis
- ☐ C) Acute tubular necrosis
- ☐ D) Bilateral hydronephrosis
- ☐ E) Rhabdomyolysis

15. A 42-year-old man with alcoholism is admitted to the hospital 12 hours after the onset of severe epigastric pain that radiates to the back. He has had several episodes of vomiting. His temperature is 38.5°C (101.3°F), pulse is 102/min, respirations are 20/min, and blood pressure is 110/70 mm Hg. There is tenderness to palpation in the epigastrium and right upper quadrant of the abdomen; there are no signs of peritonitis. Bowel sounds are decreased. After 2 days, laboratory studies show:

Hemoglobin	13 g/dL
Mean corpuscular volume	102 μm^3
Leukocyte count	12,000/mm ³
Segmented neutrophils	76%
Bands	4%
Lymphocytes	20%
Serum	
Ca ²⁺	6 mg/dL
Urea nitrogen	20 mg/dL
Creatinine	1.4 mg/dL
Albumin	3.5 g/dL
Amylase	500 U/L

His serum electrolyte concentrations are within the reference range. Which of the laboratory findings in this patient is associated with the worst prognosis?

- ☐ A) Leukocyte count
- ☐ B) Mean corpuscular volume
- ☐ C) Serum albumin concentration
- ☐ D) Serum amylase activity
- ☐ E) Serum calcium concentration



16. A 32-year-old woman comes to the emergency department because of fever and cough with increasing production of purulent sputum for 24 hours. She has had asthma since childhood that is characterized by episodes of coughing and wheezing. Treatment with inhaled corticosteroids was begun 8 months ago. She has been hospitalized twice over the past 2 months for low-grade fever, increased sputum production, and a poorly resolving infiltrate in the left lower lobe; sputum cultures at that time grew normal flora and *Aspergillus fumigatus*. She is in moderate respiratory distress. Her temperature is 38°C (100.4°F), pulse is 95/min, respirations are 22/min, and blood pressure is 110/68 mm Hg. Diffuse expiratory wheezes and crackles are heard at the left lung base. Cardiac examination shows a normal S₁ and S₂ with no murmurs. Which of the following is the most likely diagnosis?

- ☐ A) Allergic bronchopulmonary aspergillosis
- ☐ B) α_1 -Antitrypsin deficiency
- ☐ C) Cystic fibrosis
- ☐ D) Pertussis
- ☐ E) Wegener granulomatosis



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17. A previously healthy 37-year-old woman comes to the physician because of a 2-year history of progressive shortness of breath. During this period, she also has had intermittent nonproductive cough and wheezing, especially when she has colds during the winter. She has not had night sweats. She takes no medications. She has smoked one-half pack of cigarettes daily for 9 years. She is employed as a landscape architect. Her 60-year-old father has had emphysema since his early 40s. The patient appears well. Her pulse is 80/min, respirations are 16/min, and blood pressure is 120/80 mm Hg. The lungs are clear to auscultation. Breath sounds are mildly decreased at the lung bases. The remainder of the examination shows no abnormalities. Pulse oximetry on room air shows an oxygen saturation of 97%. Spirometry shows an FEV₁ of 45% of predicted. X-rays of the chest are shown. Which of the following is the most likely mechanism of this patient's lung injury?

- ☐ A) Chronic exposure to environmental allergens
- ☐ B) Chronic infection with *Aspergillus fumigatus*
- ☐ C) Chronic infection with respiratory syncytial virus
- ☐ D) Chronic irritation and inflammation from cigarette smoking
- ☐ E) Defective chloride secretion from respiratory epithelium
- ☐ F) Insufficient inhibition of neutrophil elastase

18. A 62-year-old man comes to the physician because of moderate right leg pain for 2 days. He has metastatic non-small cell carcinoma of the lung and completed palliative chemotherapy 3 weeks ago. He takes no other medications. His temperature is 37°C (98.6°F), pulse is 90/min, respirations are 22/min, and blood pressure is 146/80 mm Hg. Cardiopulmonary examination shows no abnormalities. Laboratory studies show:

Hemoglobin	13 g/dL
Platelet count	367,000/mm ³
Prothrombin time	12 sec (INR=1)
Partial thromboplastin time	28 sec
Serum creatinine	1.1 mg/dL

Venous duplex ultrasonography of the right lower extremity shows thrombosis in the common femoral vein. Which of the following is the most appropriate next step in management?

- ☐ A) Oral administration of aspirin
- ☐ B) Oral administration of warfarin
- ☐ C) Subcutaneous administration of enoxaparin
- ☐ D) CT angiography of the chest
- ☐ E) Placement of an inferior vena cava filter



19. A moderately obese 35-year-old woman has a 4-year history of recurrent abscesses in both axillae. These are progressively more frequent and severe. The episodes are interfering with her ability to work. She has type 2 diabetes mellitus that is fairly well controlled by diet and the use of oral hypoglycemic agents. Which of the following is the most appropriate management?

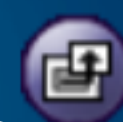
- ☐ A) Institution of insulin therapy to improve glucose control
- ☐ B) Long-term administration of antibiotics
- ☐ C) Surgical excision of apocrine tissue
- ☐ D) Topical application of silver sulfadiazine
- ☐ E) Washing three times daily with pHisoHex and water



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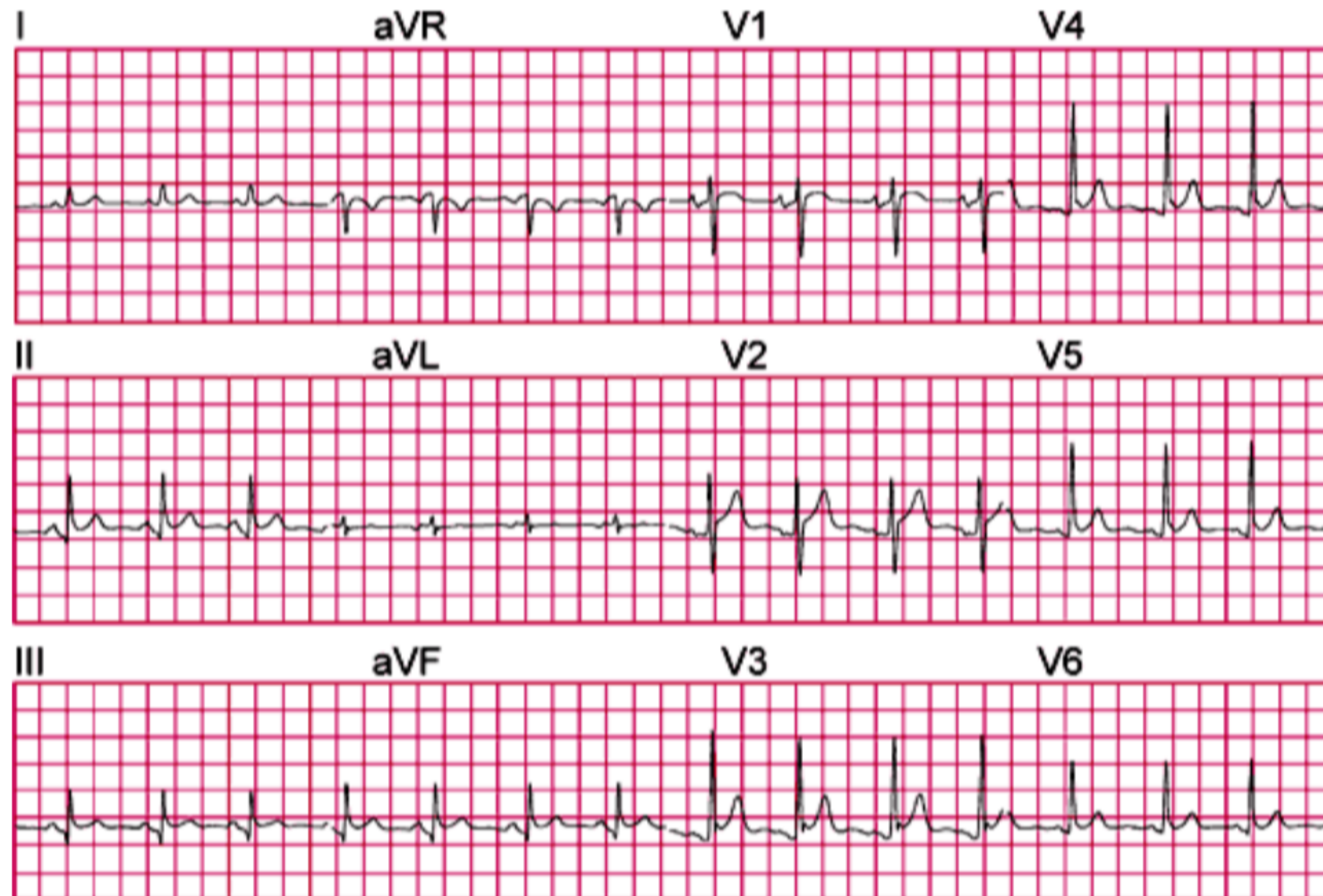
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20. A 23-year-old woman comes to the emergency department 2 hours after the sudden onset of sharp anterior chest pain. She says that the pain is worse with inspiration and when she is supine and is relieved somewhat when she sits up. She took two acetaminophen tablets 1 hour ago, but they have not relieved the pain. She has no history of similar symptoms or serious illness. She takes no other medications. Her temperature is 37.8°C (100°F), pulse is 100/min and regular, respirations are 16/min, and blood pressure is 114/76 mm Hg. A two-component friction rub is heard at the apex. The remainder of the examination shows no abnormalities. An ECG is shown. Which of the following is the most appropriate initial step in pharmacotherapy?

- ☐ A) Oral ibuprofen
- ☐ B) Oral prednisone
- ☐ C) Sublingual nitroglycerin
- ☐ D) Intravenous heparin
- ☐ E) Intravenous morphine



21. A 77-year-old woman with osteoporosis and hypertension is admitted to the hospital because of progressive confusion during the past 5 days. Her husband reports that she has had constipation for the past week and abdominal pain for the past 3 days. She had right breast cancer treated with lumpectomy and radiation therapy 2 years ago. Medications are alendronate, hydrochlorothiazide, and a calcium supplement containing vitamin D. She is alert. Her temperature is 36.4°C (97.5°F), pulse is 78/min, respirations are 20/min, and blood pressure is 156/82 mm Hg. Pulmonary examination shows no abnormalities. Abdominal examination shows mild distention and tenderness throughout; there is no guarding or rebound. Rectal examination shows hard brown stool. Neurologic examination shows no focal findings. On mental status examination, she appears anxious and is not oriented to person, place, or time. Her serum calcium concentration is 14.8 mg/dL, and serum albumin concentration is 3 g/dL. A CT scan of the head without contrast shows multiple skull metastases. Which of the following is the strongest contributing factor for this patient's hypercalcemia?

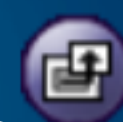
- ☐ A) Alendronate therapy
- ☐ B) Constipation
- ☐ C) Hydrochlorothiazide therapy
- ☐ D) Overuse of calcium supplement
- ☐ E) Postradiation bronchogenic carcinoma
- ☐ F) Recurrence of breast cancer



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22. A 26-year-old man is brought to the emergency department after vomiting bright red blood twice in the past hour. He appears robust and comfortable. His pulse is 116/min, and blood pressure is 100/80 mm Hg while supine. There is mild epigastric tenderness. Test of the stool for occult blood is negative. Hematocrit is 40%. Which of the following is the most appropriate next step in management?

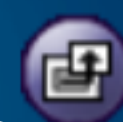
- ☐ A) Discharge him with instructions to return if the pain or vomiting recurs
- ☐ B) Repeat the hematocrit
- ☐ C) Measurement of serum amylase activity
- ☐ D) Intravenous administration of 0.9% saline
- ☐ E) Surgical consultation



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23. A 55-year-old woman comes to the emergency department because of fever, nausea, vomiting, diarrhea, and decreased urine output over the past 48 hours. She has an 8-year history of type 2 diabetes mellitus treated with glipizide. She has no history of renal disease. Her temperature is 39°C (102.2°F), and respirations are 24/min. While supine, her pulse is 106/min, and blood pressure is 116/70 mm Hg. While sitting, her pulse is 120/min, and blood pressure is 88/58 mm Hg. She appears acutely ill and slightly lethargic. Examination shows decreased skin turgor, dry mucous membranes, and flat neck veins. There is mild, generalized abdominal tenderness without guarding or rebound tenderness. Laboratory studies show:

Leukocyte count	15,000/mm ³
Serum	
Na ⁺	146 mEq/L
Cl ⁻	100 mEq/L
K ⁺	2.9 mEq/L
HCO ₃ ⁻	28 mEq/L
Urea nitrogen	72 mg/dL
Glucose	254 mg/dL
Creatinine	3.4 mg/dL
Urine	
Na ⁺	8 mEq/L
pH	5.0
Specific gravity	1.028
Glucose	2+
Ketones	trace

Which of the following is the most likely diagnosis?

- ☐ A) Acute interstitial nephritis
- ☐ B) Acute tubular necrosis
- ☐ C) Obstructive uropathy
- ☐ D) Postinfectious glomerulonephritis
- ☐ E) Prerenal azotemia



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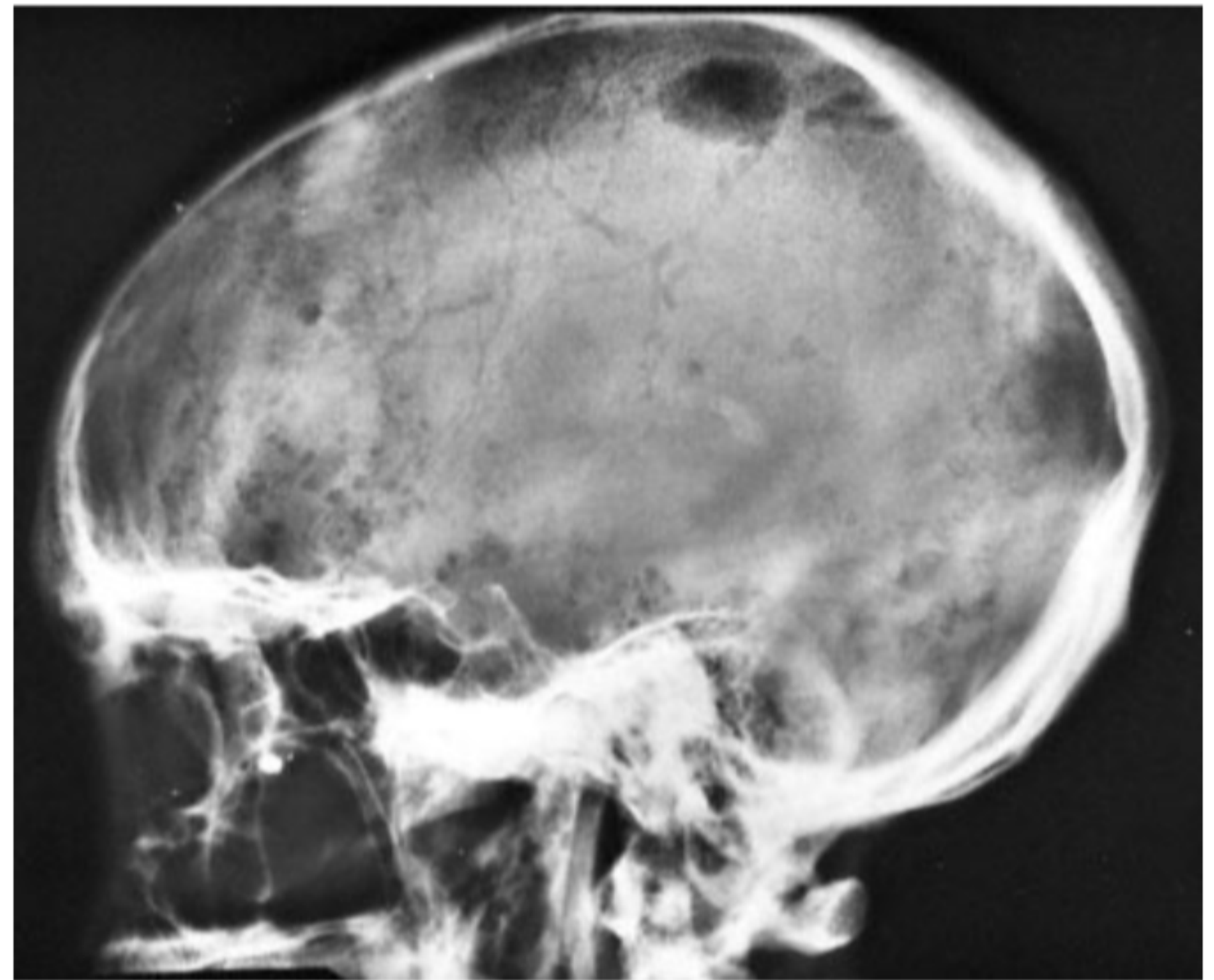


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24. A 62-year-old woman has had progressively severe midback pain over the past month. She has been fatigued and less mentally alert for 3 weeks. Her temperature is 37°C (98.6°F), pulse is 82/min, respirations are 14/min, and blood pressure is 140/80 mm Hg. The remainder of the examination shows no abnormalities except for impaired cognitive function. Laboratory studies show:

Hemoglobin	8 g/dL
Hematocrit	24%
Leukocyte count	12,500/mm ³
Serum	
Ca ²⁺	12.4 mg/dL
Urea nitrogen	35 mg/dL
Creatinine	2.4 mg/dL

Serum electrolyte concentrations are within normal limits. An x-ray of the chest shows no abnormalities. A lateral x-ray of the skull is shown. Which of the following is the most appropriate next step in evaluation?



- ☐ A) Serum protein electrophoresis
- ☐ B) Mammography
- ☐ C) Bone scan
- ☐ D) CT scan of the head
- ☐ E) Renal biopsy

25. Two days after a myocardial infarction, a previously asymptomatic 60-year-old woman develops pulmonary edema. A grade 4/6 systolic murmur is heard at the apex. Oxygen saturation in the right atrium and pulmonary artery is 65%, and pulmonary artery diastolic pressure is 35 mm Hg. Which of the following is the most likely diagnosis?

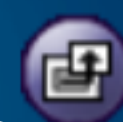
- ☐ A) Ruptured interventricular septum
- ☐ B) Ruptured left ventricular free wall
- ☐ C) Ruptured papillary muscle
- ☐ D) Ruptured pulmonary valve
- ☐ E) Torn chordae tendineae of the tricuspid valve



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26. A 32-year-old man is brought to the emergency department 30 minutes after he was found unconscious on the floor of his apartment next to an open bottle of barbiturates. On arrival, his pulse is 118/min, respirations are 6/min, and blood pressure is 98/64 mm Hg. He is intubated and mechanically ventilated. Examination shows diffuse movement to painful stimuli. Laboratory studies show:

Serum	
Na ⁺	140 mEq/L
Cl ⁻	106 mEq/L
K ⁺	6.1 mEq/L
HCO ₃ ⁻	14 mEq/L
Urea nitrogen	27 mg/dL
Creatinine	5.7 mg/dL
Creatine kinase	10,000 U/L
Urine	
pH	6
Specific gravity	1.010
Blood	positive
Glucose	negative
Protein	trace
RBC	3–4/hpf
WBC	0–2/hpf
Pigmented granular casts	4–5/lpf

Which of the following is the most likely cause of this patient's renal failure?

- ☐ A) Acute glomerulonephritis
- ☐ B) Acute interstitial nephritis
- ☐ C) Bladder outlet obstruction
- ☐ D) Cholesterol embolization
- ☐ E) Rhabdomyolysis



27. A 37-year-old woman comes to the physician because she has been amenorrheic for 3 months. She has also had increased facial hair growth and multiple episodes of acne over the past year; she has had a 27-kg (60-lb) weight gain during this time despite no change in appetite. Menses had occurred at regular intervals until 1 year ago, when they became increasingly irregular. She is not sexually active. She is 163 cm (5 ft 4 in) tall and weighs 95 kg (210 lb); BMI is 36 kg/m². Her blood pressure is 160/100 mm Hg. Examination shows truncal obesity with pink striae. The remainder of the examination shows no abnormalities. Urine β -hCG test is negative. Measurement of which of the following is the most appropriate next step in diagnosis?

- ☐ A) Serum dehydroepiandrosterone sulfate concentration
- ☐ B) Serum estradiol concentration
- ☐ C) Serum progesterone concentration
- ☐ D) Urine catecholamine concentration
- ☐ E) Urine free cortisol concentration in a 24-hour specimen



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28. A 67-year-old man with type 2 diabetes mellitus comes to the physician because of a 2-day history of increasing pain and redness of his left knee. Ten days ago, he spent several hours on his knees waxing a floor. His knee has been somewhat swollen and painful since then. His temperature is 38.1°C (100.6°F), pulse is 88/min, and blood pressure is 120/85 mm Hg. The anterior aspect of the knee overlying the patella is erythematous, and there is fluctuance in this area. There is no effusion. Range of motion of the knee is full, and any movement of the knee produces minor discomfort. Which of the following is the most likely diagnosis?

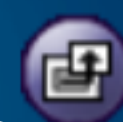
- ☐ A) Deep venous thrombosis
- ☐ B) Gout
- ☐ C) Inflamed Baker cyst
- ☐ D) Inflammatory bursitis of the pes anserinus
- ☐ E) Osgood-Schlatter disease
- ☐ F) Prepatellar septic bursitis
- ☐ G) Septic arthritis of the knee



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29. A previously healthy 24-year-old woman comes to the physician because of an itchy rash on her face, arms, and hands for 2 days. She notes that the rash appeared 2 days after working in her garden. Examination shows urticarial and vesicular lesions over the face, upper extremities, and hands. Which of the following is the most likely pathologic process of this patient's rash?

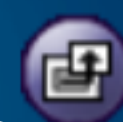
- ☐ A) Chemotaxis of eosinophils caused by direct antibody reaction with plant sap
- ☐ B) Destruction of adhesion among epidermal cells caused by circulating IgG autoantibodies directed against the cell surface of keratinocytes
- ☐ C) Inflammation caused by release of cytokines by effector and memory T lymphocytes
- ☐ D) Photosensitivity of the epidermis caused by plant sap
- ☐ E) Squamous epithelial cell damage caused by plant sap



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30. A 57-year-old man comes to the physician because of a 3-month history of mild shortness of breath with exertion and episodes of nausea and muscle pain. He received the diagnosis of type 2 diabetes mellitus 8 years ago. Current medications include glyburide and metformin. He is 180 cm (5 ft 11 in) tall and weighs 118 kg (260 lb); BMI is 36 kg/m². His blood pressure is 165/90 mm Hg. The remainder of the examination shows no abnormalities. Laboratory studies show:

Hemoglobin A _{1c}	9.2%
Serum	
Urea nitrogen	43 mg/dL
Glucose, fasting	287 mg/dL
Creatinine	2.8 mg/dL
Total bilirubin	0.8 mg/dL
AST	18 U/L
ALT	20 U/L

Which of the following is the most appropriate change to this patient's drug regimen?

- ☐ A) Add intermediate-acting insulin
- ☐ B) Add rosiglitazone
- ☐ C) Discontinue glyburide and increase the dosage of metformin
- ☐ D) Discontinue metformin and increase the dosage of glyburide
- ☐ E) Increase the dosage of glyburide and metformin
- ☐ F) Switch from glyburide and metformin to intermediate-acting insulin



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31. A 62-year-old man comes to the physician because of a 3-week history of progressive fatigue and shortness of breath with exertion. His temperature is 37°C (98.6°F), pulse is 56/min, respirations are 16/min, and blood pressure is 150/92 mm Hg. Bilateral basilar crackles are heard on auscultation. Cardiac examination shows a laterally displaced point of maximal impulse. Echocardiography shows left ventricular dilation and an ejection fraction of 32%. In addition to furosemide, which of the following is the most appropriate pharmacotherapy?

- ☐ A) Clonidine
- ☐ B) Diltiazem
- ☐ C) Lisinopril
- ☐ D) Metoprolol
- ☐ E) Nifedipine
- ☐ F) Spironolactone



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32. An asymptomatic 32-year-old woman with HIV infection comes to the physician as a new patient. Her last visit to a physician was 14 months ago. She takes no medications. Examination shows hairy leukoplakia and cervical adenopathy. Her CD4+ T-lymphocyte count is $180/\text{mm}^3$ (Normal ≥ 500), and plasma HIV viral load is 20,000 copies/mL. In addition to antiretroviral therapy, which of the following is the most appropriate next step in management?

- ☐ A) Mammography and Pap smear
- ☐ B) Mammography, Pap smear, and *Pneumocystis jiroveci* (formerly *P. carinii*) prophylaxis
- ☐ C) Pap smear and *P. jiroveci* prophylaxis
- ☐ D) *P. jiroveci* and *Mycobacterium avium* complex prophylaxis
- ☐ E) *P. jiroveci* prophylaxis only



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33. A 37-year-old woman with type 1 diabetes mellitus and anxiety disorder comes to the physician because of a 6-month history of progressive numbness of her hands that has recently extended to her forearms. During the past 2 months, she has burned her fingers numerous times while cooking; she did not know that she had burned herself until she saw the development of blisters. She has no other neurologic symptoms. Her only medication is insulin. Cranial nerves are intact. Muscle strength is 5/5 in the upper and lower extremities. Deep tendon reflexes are 3+ bilaterally. Babinski sign is absent. Sensation to pinprick and temperature is absent over the hands and upper extremities extending to the neck and posterior scalp but is intact over the trunk and lower extremities. Proprioception and sensation to vibration are normal throughout. Which of the following is the most likely diagnosis?

- ☐ A) Brachial plexopathy
- ☐ B) Cervical stenosis
- ☐ C) Conversion disorder
- ☐ D) Diabetic polyneuropathy
- ☐ E) Syringomyelia



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34. A 62-year-old man with severe obstructive pulmonary disease has chronic disabling dyspnea. His pulse is 90/min, respirations are 30/min, and blood pressure is 150/85 mm Hg. He is plethoric, and jugular veins are distended. Breath sounds are distant, and the second component of S_2 is accentuated. His FEV_1 is 0.5 L. An x-ray of the chest shows prominent pulmonary arteries. The pulmonary hypertension in this patient is most likely the result of which of the following?

- ☐ A) Decreased cardiac output
- ☐ B) Decreased expiratory flow rate
- ☐ C) Increased residual volume
- ☐ D) Severity of hypercarbia
- ☐ E) Severity of hypoxemia



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35. A 57-year-old man comes to the emergency department because of a painful, cold right leg for 4 hours. He has hypertension, type 2 diabetes mellitus, and hyperlipidemia. Medications include metformin, triamterene-hydrochlorothiazide, and atorvastatin. He has smoked one pack of cigarettes daily for 40 years. His blood pressure is 140/90 mm Hg. Pulses are decreased in the right lower extremity. His serum creatinine concentration is 1.5 mg/dL. Angiography of the right lower extremity shows occlusion of the common femoral artery. Twenty-four hours later, his serum creatinine concentration increases to 3.6 mg/dL. Which of the following measures prior to his angiography is most likely to have prevented this patient's worsening azotemia?

- ☐ A) Discontinuation of metformin
- ☐ B) Hydration
- ☐ C) Pretreatment with mannitol
- ☐ D) Pretreatment with prednisone and diphenhydramine
- ☐ E) There are no known preventive measures



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36. A 62-year-old woman comes to the physician because of increasingly severe pain of the fingers over the past 6 months. She has a 6-year history of cold sensitivity of the hands. On cold exposure, her fingers become initially pale and then turn blue; there is redness with recovery. She has a 3-year history of gastroesophageal reflux disease. Examination shows tightening of the skin of the fingers, with erythema, scaling, and telangiectasias. Dry crepitant crackles are heard at both lung bases. X-rays of the hands show loss of tufts of some distal phalanges and scattered soft tissue calcifications. Serum antinuclear antibody testing is most likely to show which of the following?

- ☐ A) Negative findings
- ☐ B) Positive findings for anticentromere antibody
- ☐ C) Positive findings for antineutrophil cytoplasmic antibody and negative findings for antinuclear antibody
- ☐ D) Positive findings for antiribonuclear protein antibody
- ☐ E) Positive findings for double-stranded DNA



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37. A 62-year-old man comes to the physician for a preoperative evaluation. He is scheduled to undergo partial colectomy in 5 days because of recurrent diverticulitis. He can carry groceries up two flights of stairs without stopping. He has no history of chest pain or heart disease. Two months ago, his serum total cholesterol concentration was 162 mg/dL. He takes no medications. He is adopted and has no information about his family history. He smoked one-half pack of cigarettes daily for 30 years but stopped 3 years ago. He is 175 cm (5 ft 9 in) tall and weighs 93 kg (205 lb); BMI is 30 kg/m². His pulse is 78/min and regular, and blood pressure is 138/84 mm Hg. The lungs are clear to auscultation. S₁ and S₂ are normal. A grade 2/6 holosystolic murmur is heard at the apex with radiation toward the left axilla. The abdomen is soft. An ECG shows a normal sinus rhythm and left bundle branch block. Echocardiography shows mitral valve thickening with mild regurgitation. Which of the following diagnostic tests is indicated prior to the operation?

- ☐ A) Exercise stress test
- ☐ B) 24-Hour ambulatory ECG monitoring
- ☐ C) Electron beam (ultrafast) CT scan
- ☐ D) Coronary angiography
- ☐ E) No further tests are indicated



38. A 52-year-old woman comes for a routine health maintenance examination. She feels well. Over the past 6 months, menses have occurred at irregular 2- to 3-month intervals. She takes no medications. She has smoked one pack of cigarettes daily for 30 years and drinks alcohol socially. Her father has hypertension, and her mother has osteoarthritis of the knees and is scheduled for bilateral knee replacements. The patient is 157 cm (5 ft 2 in) tall and weighs 72 kg (160 lb); BMI is 29 kg/m². Her pulse is 80/min and regular and blood pressure is 115/70 mm Hg. Range of motion of all joints is full and does not produce pain. She is concerned about her risk for arthritis. Which of the following is most likely to decrease this patient's risk for osteoarthritis?

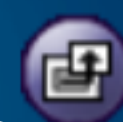
- ☐ A) Begin a swimming program 30 minutes daily three times weekly
- ☐ B) Begin a walking program 30 minutes daily three times weekly
- ☐ C) Smoking cessation
- ☐ D) Weight loss
- ☐ E) Estrogen replacement therapy
- ☐ F) Nonsteroidal anti-inflammatory drug therapy
- ☐ G) No preventive measures available



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39. A 42-year-old man comes to the emergency department because of the recent onset of shortness of breath. He has smoked two packs of cigarettes daily for 25 years. He is emaciated and in moderate respiratory distress. There is dullness to percussion and egophony at the right lung base. He has a leukocyte count of $18,200/\text{mm}^3$ (60% neutrophils, 17% band forms, 20% lymphocytes, and 3% monocytes). An x-ray of the chest shows a right lower lobe infiltrate with a pleural effusion. Thoracentesis shows purulent, foul-smelling fluid. Which of the following is the most likely diagnosis?

- ☐ A) Anaerobic empyema
- ☐ B) Chylothorax
- ☐ C) Malignant effusion
- ☐ D) Pneumococcal empyema
- ☐ E) Tuberculous empyema



40. A 62-year-old man is admitted to the hospital for treatment of pneumonia. He has a 3-day history of fever, cough productive of greenish sputum, and mild shortness of breath. He has type 2 diabetes mellitus and coronary artery disease. Current medications include 81-mg aspirin, atenolol, isosorbide, glipizide, metformin, and rosiglitazone. He appears ill and has mild shortness of breath. His temperature is 39°C (102.2°F), pulse is 110/min and regular, respirations are 24/min, and blood pressure is 195/64 mm Hg. Moist crackles are heard at the left lung base. Laboratory studies show:

Leukocyte count	16,700/mm ³
Segmented neutrophils	90%
Bands	2%
Lymphocytes	8%
Serum	
Na ⁺	137 mEq/L
K ⁺	4.8 mEq/L
Cl ⁻	97 mEq/L
HCO ₃ ⁻	20 mEq/L
Glucose	180 mg/dL
Creatinine	1.5 mg/dL
AST	18 U/L
ALT	12 U/L

An x-ray of the chest confirms left lower lobe pneumonia. Immediate discontinuation of which of the following is the most appropriate next step?

- ☐ A) Aspirin
- ☐ B) Atenolol
- ☐ C) Glipizide
- ☐ D) Isosorbide
- ☐ E) Metformin
- ☐ F) Rosiglitazone



41. A 72-year-old woman comes to the physician because of a 6-day history of increasingly severe left hip pain. She now has difficulty walking because of the pain. She has temporal arteritis and hypertension. Her medications are prednisone and enalapril. Vital signs are within normal limits. Internal and external range of motion of the left hip is limited by pain. The remainder of the examination shows no abnormalities. An x-ray of the left hip shows no abnormalities. An MRI of the hips is shown. Which of the following is the most likely diagnosis?

- ☐ A) Avascular necrosis
- ☐ B) Hip fracture
- ☐ C) Iliotibial band syndrome
- ☐ D) Osteoarthritis
- ☐ E) Trochanteric bursitis





42. A study is conducted to compare a new stool test with mucosal biopsy for diagnosing *Helicobacter pylori* infection. A total of 400 patients with suspected *H. pylori* infection are enrolled. Patients are screened using both methods. Results show that of the 100 patients who tested positive for *H. pylori* infection on biopsy, 80 also tested positive using the new stool test; 120 patients who tested negative for *H. pylori* infection on biopsy tested positive using the new stool test. Which of the following represents the sensitivity of the new stool test for *H. pylori* infection in this study?

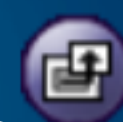
- ☐ A) 10%
- ☐ B) 20%
- ☐ C) 40%
- ☐ D) 60%
- ☐ E) 80%



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43. A 47-year-old woman comes to the physician because of a 2-day history of soreness in the back of her throat and pain on swallowing. She has had recurrent episodes of tonsillitis during the past 2 years. Examination shows massive edema of the soft palate and enlarged tonsils with a purulent discharge from the right tonsil. Without treatment, this patient is at increased risk for which of the following?

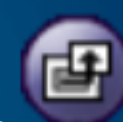
- ☐ A) Airway compromise
- ☐ B) Brain abscess
- ☐ C) Necrosis of the tonsils
- ☐ D) Paralysis of the facial nerve
- ☐ E) Rupture of the tympanic membrane



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44. A previously healthy 27-year-old intern comes to the emergency department 5 minutes after she was stuck in the left middle finger by a 20-gauge needle used on a 22-year-old patient with AIDS. The patient is currently being treated with zidovudine (AZT), indinavir, and lamivudine (3TC); his current plasma HIV viral load is undetectable. The patient's serum studies 2 days ago showed:

Total bilirubin	0.8 mg/dL
AST	100 U/L
ALT	80 U/L
Hepatitis B surface antibody	positive
Hepatitis B surface antigen	negative

The intern had completed hepatitis B immunizations 10 months ago. Examination shows a needle puncture wound. In addition to chemoprophylaxis for HIV exposure, which of the following is the most appropriate intervention for this intern?

- ☐ A) Washing the area of involvement
- ☐ B) Administering hyperimmune hepatitis B immune globulin
- ☐ C) Administering immune globulin
- ☐ D) Administering interferon alfa
- ☐ E) No intervention is required



45. A 72-year-old man comes to the physician for a routine examination. He has a 1-year history of hypertension controlled by a thiazide diuretic. He smokes one pack of cigarettes and drinks 3 ounces of alcohol daily. He has gained 6 kg (13 lb) since his last visit 1 year ago. He is 178 cm (5 ft 10 in) tall and now weighs 88 kg (195 lb); BMI is 28 kg/m². His blood pressure is 156/88 mm Hg. His serum LDL-cholesterol concentration is 150 mg/dL, and serum HDL-cholesterol concentration is 50 mg/dL. Which of the following is the greatest risk factor for coronary artery disease in this patient?

- ☐ A) Alcohol consumption
- ☐ B) Blood pressure
- ☐ C) Cigarette smoking
- ☐ D) Use of a thiazide diuretic
- ☐ E) Weight



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46. A 24-year-old woman, gravida 2, para 2, comes to the emergency department because of an uncontrollable nosebleed for 6 hours. This is her fourth nosebleed over the past 2 weeks. Each of the previous nosebleeds stopped within approximately 2 hours. Her menses occur at regular 28-day intervals, but she did have heavier than usual bleeding during her last two menstrual periods. She takes no medications. She has no history of bleeding problems, including during an uncomplicated appendectomy at the age of 10 years and during delivery of her two children. There is no family history of bleeding problems. Examination shows scattered petechiae over the trunk and upper and lower extremities. There is fresh and clotted blood in the nares with continued oozing of blood. Her hemoglobin concentration is 9.8 g/dL, leukocyte count is $9000/\text{mm}^3$ with a normal differential, and platelet count is $20,000/\text{mm}^3$. Examination of bone marrow aspirate shows megakaryocytes. Which of the following is the most appropriate next step in management?

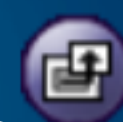
- ☐ A) Corticosteroid therapy
- ☐ B) Methotrexate therapy
- ☐ C) Splenic artery embolization
- ☐ D) Splenic irradiation
- ☐ E) Splenectomy



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47. A 77-year-old man is brought by his wife for a follow-up examination 3 weeks after an episode of right middle lobe pneumonia. The pneumonia is improving with antibiotic therapy. Three months ago, he had a similar episode. His wife says that he swallows his food well, but he has had a slight increase in his chronic cough after meals. He has multi-infarct (vascular) dementia. He smoked two packs of cigarettes daily for 40 years, but he quit 15 years ago. He has never had chronic obstructive pulmonary disease. Rhonchi are heard over the right midlung field on auscultation; the diaphragm has poor excursion. His speech is mildly dysarthric. His Mini-Mental State Examination score is 20/30. Which of the following is the most appropriate next step in management?

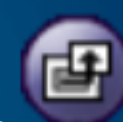
- ☐ A) Recommendation of a full liquid diet with liquid supplements
- ☐ B) Swallowing study
- ☐ C) CT scan of the chest
- ☐ D) Bronchoscopy
- ☐ E) Inhaled bronchodilator therapy



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48. A healthy 39-year-old man of German descent comes for a routine examination. He had a cholecystectomy at the age of 25 years. He has had several episodes of mild jaundice over the past 10 years that resolved spontaneously. He appears healthy. Examination shows no abnormalities except for splenomegaly. His hematocrit is 39%, mean corpuscular hemoglobin concentration is 36% Hb/cell, and mean corpuscular volume is $80 \mu\text{m}^3$. Direct antiglobulin (Coombs) test is negative. A blood smear shows erythrocytes with loss of central pallor. Which of the following is the most likely cause of this patient's anemia?

- | | |
|--|---|
| ☐ A) Abnormal red blood cell membrane | ☐ F) Globin chain mutation |
| ☐ B) Autoimmune hemolytic anemia | ☐ G) Glucose 6-phosphate dehydrogenase deficiency |
| ☐ C) Congenital dihydrofolate reductase deficiency | ☐ H) Lead poisoning |
| ☐ D) Decreased erythropoietin concentration | ☐ I) Pyruvate kinase deficiency |
| ☐ E) Folic acid deficiency | ☐ J) Vitamin B ₁₂ (cobalamin) deficiency |



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49. A 47-year-old man comes to the physician because of a 1-month history of progressive swelling of both legs and generalized weakness. He says he now has some difficulty performing his usual activities. Last year, he received the diagnosis of hepatitis B and C. He has a long-standing history of intravenous drug use. He refused treatment for hepatitis C and currently takes no medications. He is not in acute distress. He is 183 cm (6 ft) tall and weighs 55 kg (121 lb); BMI is 16 kg/m². His temperature is 37°C (98.6°F), pulse is 70/min, respirations are 12/min, and blood pressure is 160/99 mm Hg. Cardiopulmonary examination shows no abnormalities. The abdomen is nontender. There is no hepatomegaly. Muscle strength is normal. Laboratory studies show:

Hemoglobin	10 g/dL
Leukocyte count	9000/mm ³
Platelet count	260,000/mm ³
Serum	
Urea nitrogen	22 mg/dL
Creatinine	2.1 mg/dL
Albumin	3.8 g/dL
Total bilirubin	0.6 mg/dL
AST	70 U/L
ALT	60 U/L
Urine	
Blood	3+
Protein	4+

Microscopic examination of urine shows numerous RBCs, which are mostly acanthocytes with irregular membrane protrusions. Which of the following is the most appropriate next step in diagnosis?

- ☐ A) Second set of laboratory studies in 1 month
- ☐ B) Urine protein electrophoresis
- ☐ C) Urine culture
- ☐ D) Cystoscopy
- ☐ E) Renal biopsy



50. A 37-year-old man comes to the emergency department because of vomiting for the past 3 days. He says that he cannot hold down liquids or solids. For the past month, he has had early satiety and has not been eating as much as he usually does. He has a 2-year history of peptic ulcer disease, diagnosed by upper endoscopy. Infection with *Helicobacter pylori* was successfully treated with clarithromycin, amoxicillin, omeprazole, and bismuth subsalicylate therapy. Since then, he has had intermittent epigastric pain for which he takes cimetidine when he remembers to do so. He appears ill and anxious. His temperature is 37°C(98.6°F), pulse is 100/min, respirations are 14/min, and blood pressure is 110/60 mm Hg. Physical examination shows dry mucous membranes and a flat, soft, nontender abdomen. Which of the following sets of laboratory abnormalities is most likely in this patient?

- | | K ⁺ | Cl ⁻ | pH |
|--------------------------|----------------|-----------------|-----------|
| ☐ A) | Increased | increased | decreased |
| ☐ B) | Decreased | decreased | decreased |
| ☐ C) | Increased | decreased | increased |
| ☐ D) | Decreased | decreased | increased |
| ☐ E) | Decreased | increased | increased |

